

SpaceX Falcon 9 Launch Analysis and Predictive

IBM Applied Data Science Capstone

Executive Summary

This project analyzes historical Falcon 9 launch data to understand launch-site performance, payload and mission patterns, and first-stage landing outcomes. Data collection, wrangling, exploratory analysis, SQL, interactive visual analytics, dashboard development, and machine learning are combined into an end-to-end data science workflow.

The predictive component treats first-stage landing success as a classification problem and compares multiple machine learning approaches.

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Introduction

- Falcon 9 is a reusable launch vehicle, making first-stage recovery an important part of launch economics.
- The project studies historical launch data to identify patterns in launch activity and landing success.
- The analysis focuses on launch sites, payload characteristics, mission outcomes, launch trends, and landing outcomes.
- The final goal is to use available launch features to build a predictive model for first-stage landing success.

Data Collection Methodology

- SpaceX REST API was used to collect historical launch information.
- Web scraping was used to collect additional historical launch information.
- The collected data was normalized into analysis-ready tabular structures.
- Relevant fields include launch date, launch site, payload information, orbit, booster information, mission outcome, and landing outcome.

Data Wrangling Methodology

- Missing and inconsistent values were handled during preprocessing.
- Relevant variables were selected for exploratory and predictive analysis.
- Categorical and numerical variables were converted into suitable formats.
- A binary target representing first-stage landing success was prepared for classification.
- The cleaned data was used consistently across EDA, SQL, visualization, and machine learning tasks.

Exploratory Data Analysis Methodology

- Launch outcomes were summarized to understand overall mission performance.
- Launch sites were compared by launch frequency and outcomes.
- Payload characteristics were examined against launch and mission outcomes.
- Launch activity was analyzed across time.
- Categorical and numerical relationships were explored using Python visualization libraries.

EDA with Visualization: Results

- Launch activity shows how Falcon 9 operations changed over time.
- Launch-site distributions reveal that activity is concentrated at a small number of major sites.
- Launch outcome visualizations distinguish successful and unsuccessful missions.
- Payload and launch characteristics provide additional context for understanding mission outcomes.

EDA with SQL: Methodology

- SQL queries were used to filter, aggregate, group, and summarize the launch dataset.
- The analysis examined payload mass, launch-site activity, and mission outcomes.
- Grouped queries were used to compare mission outcomes across launch sites.
- SQL results were used as a complementary analytical view alongside Python-based EDA.

EDA with SQL: Results

- The SQL analysis identified the launch sites represented in the dataset.
- Payload-mass queries were used to determine minimum and aggregate payload characteristics.
- Mission outcomes were counted by launch site to compare site-level activity.
- The analysis provides a structured view of launch-site and payload patterns using relational queries.

Interactive Map with Folium

- Folium was used to create an interactive geographic visualization of SpaceX launch sites.
- Markers represent launch locations and allow geographic exploration.
- Launch-site information can be inspected interactively.
- The map connects launch performance analysis with the physical location of launch facilities.

Folium: Interactive Visual Analytics Results

- The interactive map makes the geographic distribution of launch sites immediately visible.
- Major Falcon 9 launch locations can be compared spatially.
- Geographic context helps explain differences in launch activity between sites.

Plotly Dash: Methodology

- A Plotly Dash application was designed for interactive launch analysis.
- A launch-site dropdown enables users to filter the analysis.
- Interactive Plotly charts display launch outcomes and launch trends.
- The dashboard provides a user-facing interface for exploring launch performance.

Plotly Dash: Results

- The dashboard enables launch-site-specific exploration.
- Success and failure outcomes can be viewed interactively.
- Yearly launch trends provide a time-based view of activity.
- Interactive filtering makes the analysis easier to explore than static charts alone.

Predictive Analysis: Methodology

- First-stage landing success was formulated as a binary classification problem.
- The data was divided into training and testing sets.
- Features were prepared and standardized where appropriate.
- Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors were evaluated.
- Accuracy was used as a primary model-comparison metric.

Predictive Analysis: Results

- The trained classification models were evaluated on held-out test data.
- Model accuracies were compared to identify the strongest candidate.
- A confusion matrix was used to inspect correct and incorrect classifications.
- A classification report provided precision, recall, and F1-score information.
- The predictive workflow demonstrates how launch data can be used to study first-stage landing success.

Key Insights and Innovative Perspective

- Falcon 9 launch activity can be studied from operational, geographic, and predictive perspectives.
- Launch-site concentration provides a useful operational view of the program.
- Reusable first-stage landing success is an important outcome to analyze separately from overall mission success.
- Combining geographic visualization, interactive dashboards, SQL, and machine learning creates a more complete analytical workflow.

Conclusion

The project demonstrates a complete data science pipeline from data collection to predictive analysis. SpaceX Falcon 9 launch data was collected, cleaned, explored with Python and SQL, visualized geographically and interactively, and used to develop classification models for first-stage landing success.

The resulting workflow provides both descriptive insights into launch operations and a predictive framework for studying landing outcomes.

Software and Frameworks

Area	Technologies
Programming	Python
Data Collection	Requests, BeautifulSoup, SpaceX REST API
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
SQL	SQLite / SQL
Geospatial	Folium
Dashboard	Plotly Dash
Machine Learning	Scikit-learn
Environment	Jupyter Notebook

Thank You

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